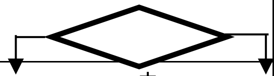
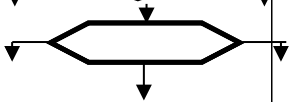
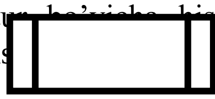
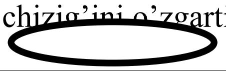
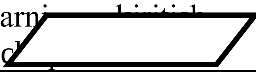
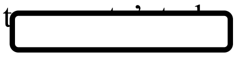
AMALIY MASHG'ULOT-8

Mavzu: Axborot jarayonlarini algoritmlash va dasturlash. Algoritmlash asoslari. Zamonaviy dasturlash tillari va texnologiyalari.

Mashg'ulot maqsadi: Talabalarda C++ Builderda chiziqli, tarmoqlanuvchi va takrorlanuvchi jarayonlarga algoritmlar tuzish.
2. Bu bo'limda chiziqli, tarmoqlanuvchi va takrorlanuvchi algoritmga keltiriladigan masalalarni Delphi va Borland C++ da yechish bayon etilgan.

1.Talabalarda axborotlashtirish jarayoni to'g'risida umumiy va asosiy bilimlarni hosil qilish.
2. Algoritm va algoritmning berilishi usullarito'g'risida mukammal tasavvurga ega bo'lish.
3. Algoritmning turlari va chiziqli algoritmlar tuzish bo'yicha ko'nikmalarni shakllantirish. Algoritmlarni ko'rgazmaliroq qilib tasvirlash uchun blok-sxema, ya'ni geometrik usul ko'proq qo'llaniladi. Algoritmning blok-sxemasi algoritmning asosiy tuzilishining yaqqol geometrik tasviri: algoritm bloklari, ya'ni geometrik shakllar ko'rinishida, bloklar orasidagi aloqa esa yunaltirilgan chiziqlar bilan ko'rsatiladi. Chiziqlarning yunalishi bir blokdan so'ng qaysi blok bajarilishini bildiradi. Algoritmlarni ushbu usulda ifodalashda vazifasi, tutgan o'rniga qarab quyidagi geometrik shakl (blok) lardan foydalaniladi.

Blokning atalishi	Belgilanishi	Tushunilishi
Hisoblashlar bloki (to'g'ri-to'rtburchak)		Hisoblash amali yoki hisob-lash amallari ketma-ketligi
shartli blok (romb)		Shartlarni tekshirish
siklik jarayon (oltiburchak)		Siklning boshlanishi
qism dastur		qism dastur bo'yicha hisoblash, standart qis
birlashtirish (aylana)		Yo'nalish chizig'ini o'zgarti-rish
Ma'lumotlarni kiritish va chiqarish (parallelogramm)		Ma'lumotlarni kiritish va natijalarni ch
Algoritmning boshi va oxiri (oval)		Boshlash, t
Chiqarish bloki		Ma'lumotlarni qog'ozga chiqarish

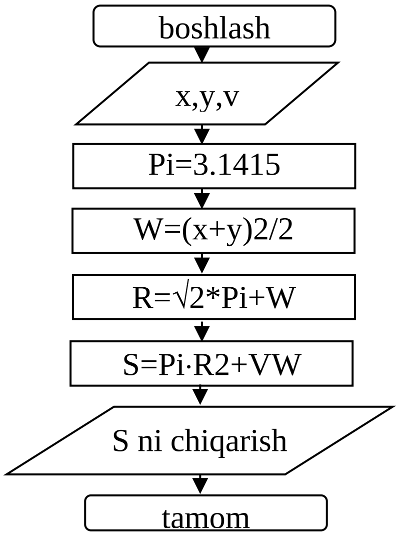
Topshiriqlarni bajarilish namunasi
1-vazifa

Chiziqli algoritm

Ifodaning qiymatini hisoblash algoritmi (blok sxema) tuzing.

$S = \pi R^2 + VW$, bu yerda $W = \dfrac{x+y}{2}$; $R = \sqrt{2\pi + W}$

a) Masalani yechish algoritmi (blok-sxema).



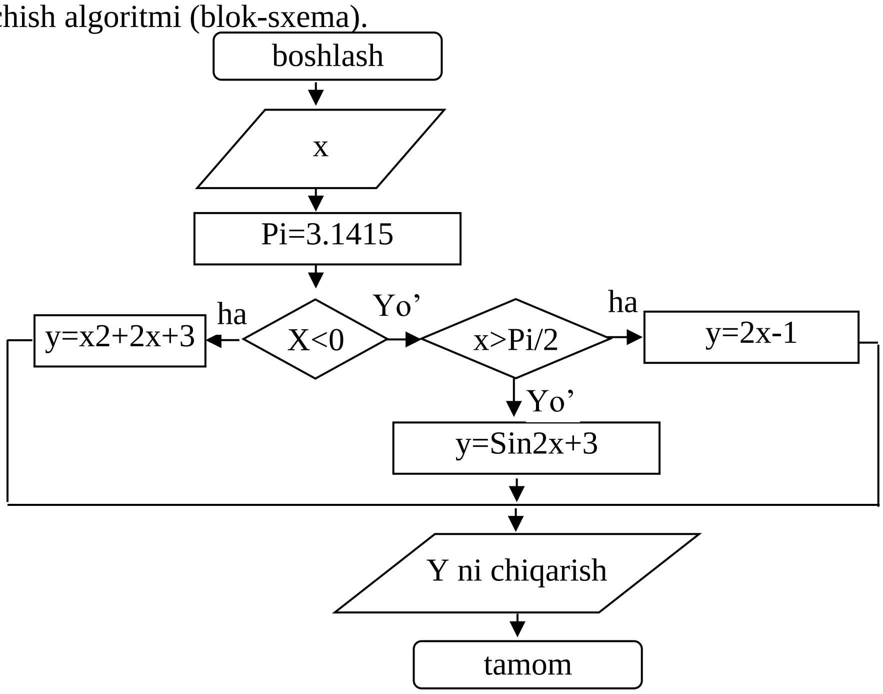
2–vazifa

Tarmoqlanuvchi algoritm

Ifodaning qiymatini hisoblash algoritmi (blok-sxema) tuzing.

$$y = \begin{cases} x^2 + 2x + 3 \text{ , } agar \ x < 0 \\ 2x - 1, \qquad agar \ \ x > \dfrac{\pi}{2} \\ \sin^2 x + 3 \text{ , } agar \ \ 0 \le x \le \dfrac{\pi}{2} \end{cases}$$

a) Masalani yechish algoritmi (blok-sxema).



3–vazifa

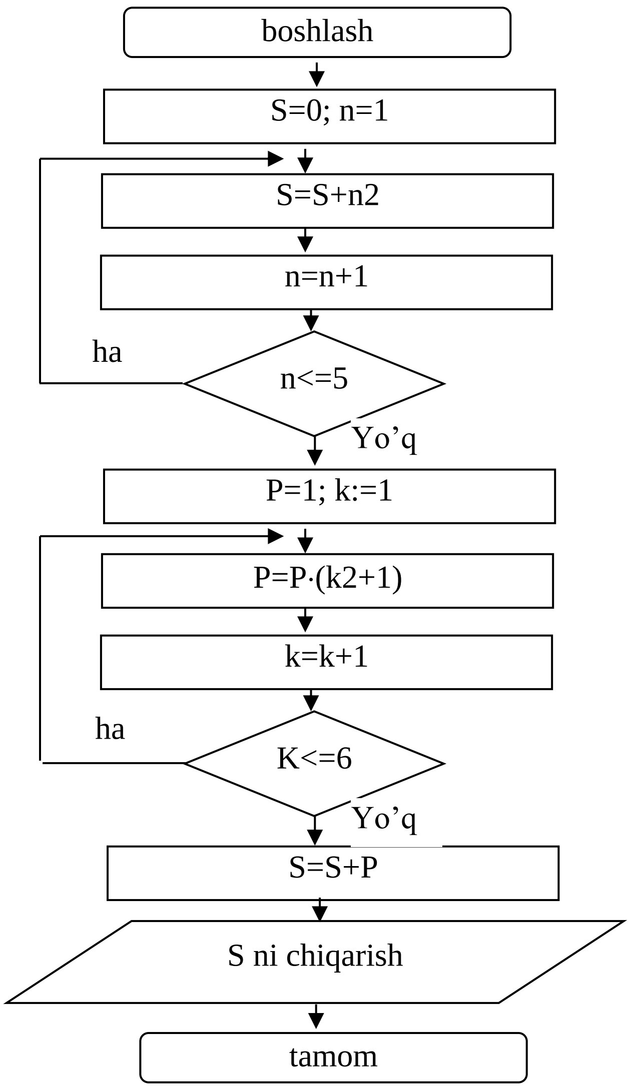
Takrorlanuvchi jarayonlarga algoritm (blok-sxema) tuzish.

Topshiriqlarni bajarish namunasi

a) Ifodaning qiymatini hisoblash algoritmi (blok-sxema) tuzing.

$$S = \sum_{n=1}^{5} n^2 + \prod_{k=1}^{6} (k^2 + 1)$$

1) Masalani yechish algoritmi (blok-sxema).

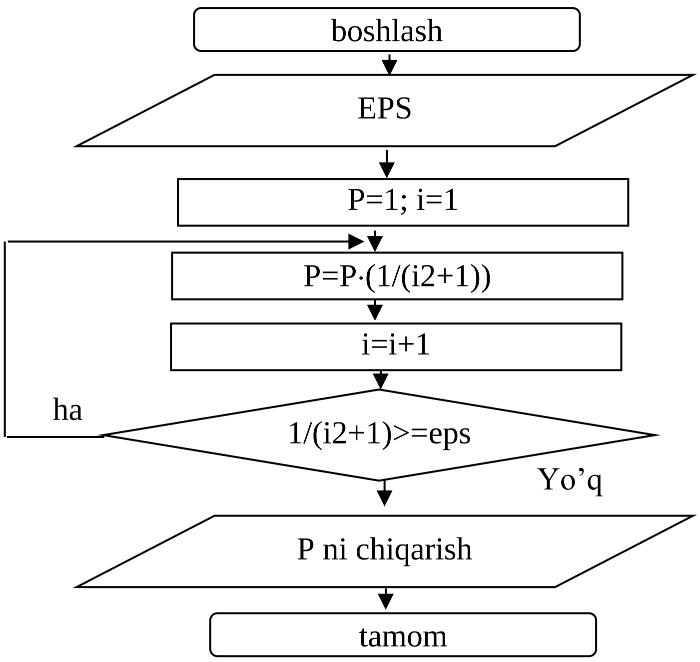


4– vazifa

b)Sharti oldin yoki sharti keyin qo‘yilgan sikl operatoridan foydalanib quyidagi ifodaning qiymatini eps aniqlik bilan hisoblash algoritmini tuzing.

$$P = \prod_{i=1}^{\infty} \frac{1}{i^2 + 1}\ ,$$ bu yerda eps = 0,001.

1) Masalani yechish algoritmi (blok-sxema).

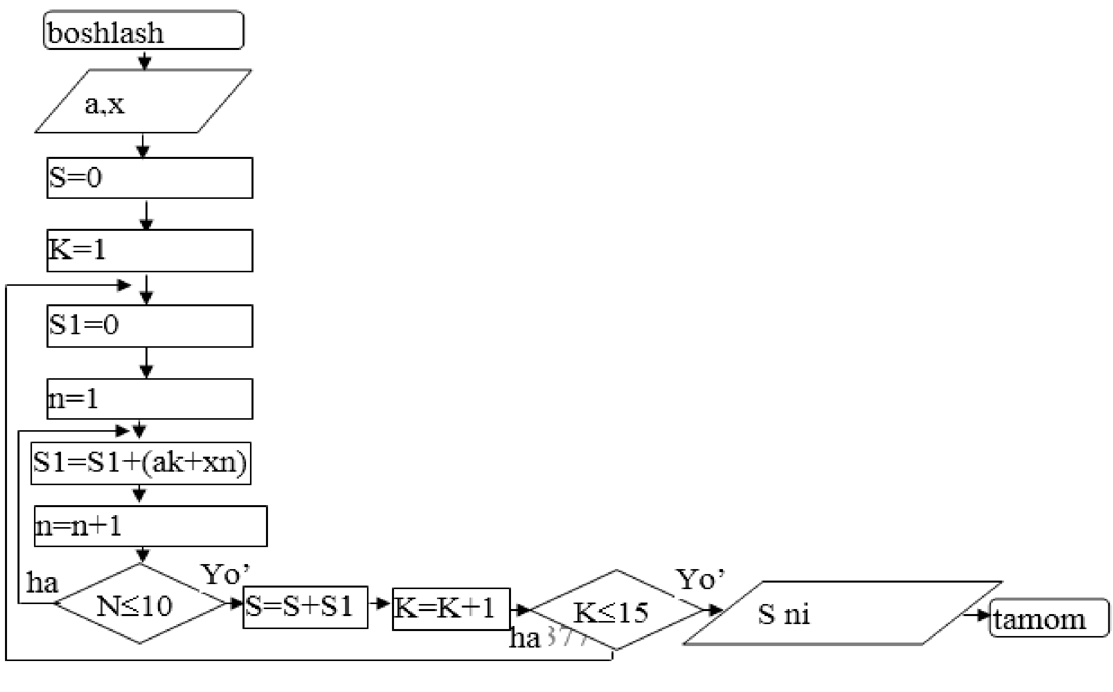


5– vazifa

c) Ichma-ich joylashdan sikllardan foydalanib ifoda qiymatini hisoblash algoritmi tuzing:

$$S = \sum_{k=1}^{15} \sum_{n=1}^{10} (a^k + x^n)$$

1) Masalani yechish algoritmi (blok-sxema).



1-Amaliy mashgulot uchun topshiriqlar.

Amaliy mashg'ulot topshiriqlarni bajarishda har bir talaba dastlab chiziqli, tarmoqlanish va tanlash buyruqlari haqida qisqacha ma'lumot berib, so'ngra berilgan amaliy va laboroto'riya tooshiriqlarni bajaridi.

1-variant

a) $a = u^{(x+y)/2} - \sqrt{\dfrac{x-1}{y+1}}$ $b = \sin(2\arccos t)$, **bu yerda** $x = 12{,}65$; $y = -2{,}255$; $u = 3{,}205$, $t = 0{,}88$

2-variant

a) $a = (2 + y^2)\dfrac{x + y/2}{y^2 + 1/(1 + y^2)}, \quad b = \sqrt{\sin v + \cos t}$ **bu yerda** $x = 0{,}222$,

3-variant

a) $a = \dfrac{1}{2}\left(x^{y-x} + y^{(x+y)/2}\right), \quad b = \lg\left(\sqrt[3]{u} + \sqrt{v+2}\right)$, **bu yerda** $x = 3{,}225, \quad y = 2{,}98, \quad u = 125{,}33, \quad v = 33{,}075$

4-variant

a) $a = \sqrt[3]{x + \sqrt[4]{y}}; \quad b = \sqrt{y}\, l^{-(y+4/2)}$, **bu yerda** $x = 37{,}15; \; y = 12{,}55; \quad l = 20{,}12.$

5-variant

a) $a = \sqrt{\left|\cos^2 x^2 + y^2\right|}$, $b = \dfrac{e^{3xy} + \ln\left|3x^2 - y\right|}{\sqrt{x^2 + y^2}}$, **bu yerda x=1,42, y=2,035**

6-variant

a) [a,b] oraliqdagi m soniga karrali sonlar yig'indisini hisoblang.

$S = \sum_{k=1}^{\infty} \dfrac{k^2}{x^k}$ **ni eps = 0,001 aniqlik bilan hisoblang.**

7-variant

a) $a = \dfrac{(x+y)^3 + \sqrt[3]{|x+y|}}{3}$, $b = \dfrac{\ln\left|\frac{x}{y}\right| + \lg\left|\frac{y}{x}\right|}{e^{xy}}$ **; bu yerda x=1,0645, y=2,1365.**

8-variant

$z = (1+u)\dfrac{x + \dfrac{y}{u}}{a - \dfrac{1}{1-x^2}}$; $u = \dfrac{\sin^2 x}{x^2 + y^2}$; $x = \dfrac{1}{2}$ **; a=10,5, y=2,5 .**

9-variant

a) $t = e^{|x|} \cdot tg\dfrac{a+b}{a-b}, \quad b = \dfrac{\cos x + \sin x^2}{e^x + a},$ **bu yerda a=10, v=5, x=2.**

$$Z = \begin{cases} \sin \ln|x| + \sqrt{|x|^3 + 5{,}3}, & agar \quad x < -3 \\ x^2 + e^{|x|} + tgx, & agar \quad x = -3 \\ 0{,}8 \cdot \sin x + 10^{-5{,}6}, & agar \quad x > -3 \end{cases}$$

10-variant

a) $l = tg4{,}5x + \dfrac{\left|x^{-10}\right| + b}{\sin 0{,}5x + a}, \quad t = \dfrac{e^{a+b} + \sqrt{x^2 + 3}}{a}$ **, bu yerda a=10, b=16, x=-2,5.**

11-variant

a) $a = \dfrac{y^{x+1}}{\sqrt[3]{y-2} + 3} + \dfrac{x + y/2}{2x + y}; \quad b = (x-1)^{-1/\sin^2 z};$

$$x = 1{,}625; \quad y = 15{,}4; \quad z = 0{,}255$$

12-variant

a) $a = \dfrac{x^{y-z} + e^{y-1}}{1 + xy - tgz}; \quad b = 1 + y - \dfrac{y - x^2}{2} + \dfrac{(x-y)^3}{3};$

$$x = 2{,}444; \quad y = 0{,}869; \quad z = -0{,}166.$$

13-variant

a) $a = \lg\left(\sqrt{e^{x-y}} + x^y + z\right); \quad b = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!};$

$$x = 1{,}542; \quad y = -3{,}264; \quad z = 8{,}05$$

14-variant

a) $a = \dfrac{\sqrt[3]{8 - x - y^2 + 1}}{x^2 + y^2 + 1}; \quad b = e^{x-y}\left(tg^2 z + 1\right);$

$$x = 4{,}5; \quad y = 0{,}007; \quad z = 0{,}845$$

15-variant

a) $a = \cos x + \cos y^{1-2\sin^2 y}; \quad b = 1 + z + \dfrac{z^2}{2} + \dfrac{z^3}{3} + \dfrac{z^4}{4};$

$$x = 0{,}4; \quad y = -0{,}875; \quad z = -0{,}475$$

b)

16-variant

a) $a=\sqrt{10\left(\sqrt[3]{x}+x^{y+2}\right)}$; $\quad b=\left(\arcsin z\right)^2+x+y$;

$$x=16{,}55;\quad y=-2{,}75;\quad z=0{,}15$$

17-variant

a) Uchlarining koordinatalari bilan berilgan uchburchakning perimetri va yuzasini toping.

$$z=\begin{cases}\frac{1}{x}+\frac{1}{y}+\frac{1}{(x+y)}, & agar \quad x\neq 0 \ \ va \ \ y\neq 0\\ \frac{1}{x}+\frac{1}{(x+1)}, & agar \quad x\neq 0 \ \ va \ \ y=0\\ \frac{1}{y}+\frac{1}{(y+1)}, & agar \quad x=0 \ \ va \ \ y\neq 0\end{cases}$$

18-variant

a) $S=2\pi R^3+\sqrt{R+vt}$, $\quad t=\dfrac{h+a+c}{2}$; $\quad R=\sqrt{t+2h}$

19 - variant

a) $a=u^{(x+y)/2}-\sqrt{\dfrac{x-1}{y+1}}$, $\quad b=\sin\left(2\arccos t\right)$

$$x=12{,}65;\quad y=-2{,}255;\quad u=3{,}205;\quad t=0{,}88$$

20 - variant

a) $a=\dfrac{1}{2}\left(x^{y-x}+y^{(x+y)/2}\right)$; $\quad b=\lg\left(\sqrt[3]{u}+\sqrt{v+2}\right)$,

$$x=3{,}225;\quad y=2{,}98;\quad u=125{,}33;\quad v=33{,}075$$

b) $S=\max(x,y,z)+\min(a,v,c)$

21 - variant

a) $y=a*e^{\lg x}-\sqrt{ax+b}$; $Z=t^2*\lg(|a+b|)+|x^3-2x|$,

bu yerda a=3,34; b=-2,18; x=1,128; t=3,028.

c) Berilgan N sonining barcha bo‘luvchilarini toping.

d) $P=\prod_{k=1}^{\infty}\dfrac{k}{2+k^k}$ **ni** $E=0{,}001$ **aniqlik bilan hisoblang.**